

Speed and Spillovers: Market Reaction to Pfizer's 2020 Vaccine Announcement

By Tanisha Gupta, Martin Yang, Huiya Zhang, Un Ieng Sit, Yipching Yang

I. Abstract

On November 9, 2020, Pfizer released pre-market Phase 3 interim results for its COVID-19 vaccine, creating an unambiguous, time-stamped information shock (Pfizer, 2020). This paper asks two questions: How quickly did equity markets incorporate the news? And did technologically proximate firms experience spillovers?

The study uses CRSP daily returns and a standard market model. Alphas and betas are estimated over -250 to -11 trading days; abnormal returns and cumulative abnormal returns are computed over -10 to $+10$. To gauge immediacy, the analysis also checks the first hour of trading on the event day, from the open to 10:30 a.m. ET. The sample focuses on Pfizer (PFE), Moderna (MRNA), and AstraZeneca (AZN).

Markets reacted quickly and unevenly. Pfizer's first-hour raw return was $\approx +13\%$; the day-0 abnormal return (AR) was $\approx +13.1\%$ (from CAR jumping -0.0619 on day -1 to 0.06895 on day 0). Pooled across firms, mean CAR rose to roughly $+10\%$ after the announcement. Reported 'statistical significance' refers to baseline t-tests on CARs; the intraday first-hour jump is described descriptively.

Several robustness checks support the main results in principle. The paper considers replacing the broad market with biotech benchmarks such as NBI or XBI and adding a simple two-factor version; testing narrower and wider windows— $[-1, +1]$, $[0, +1]$, $[-2, +5]$, and $[0, +20]$; and applying adjusted test statistics, bootstrapped p-values, intraday winsorization, and placebo events. The conclusions from the baseline are consistent with semi-strong efficiency: credible news is embedded in prices within minutes, and spillovers are stronger for close technology peers. Timing and clear communication appear to amplify market impact.

II. Introduction & Background

Market efficiency is often discussed in abstractions such as information sets, conditional expectations, and martingales, but it becomes tangible when a piece of news arrives at a precise time and investors have to decide what it means (Fama, 1991). Pfizer's pre-market Phase-3 interim results on November 9, 2020 are such a moment. The announcement was unambiguous and economically significant, and its release before the bell allows a direct test of price adjustment at the market open. We aim to use that natural experiment to examine two questions that travel well beyond the specific episode. How fast do prices

absorb credible corporate news, and how do those effects spill over to firms that share the relevant technology?

These questions matter because the pandemic compressed years of scientific and commercial uncertainty into a few months. Investors were not only valuing the cash flows of one company; they were also updating a shared mental model of what mRNA vaccines could do at scale. Event studies are well-suited to capturing that kind of learning because they allow us to separate firm-specific movements from broader market co-movements and to cumulate the abnormal piece into a digestible statistic (MacKinlay, 1997; Kothari & Warner, 2007). The design we adopt is intentionally simple: a market model estimated on a long pre-event window, abnormal returns calculated in a short window around the announcement, and CARs as the main summary. We keep the focus on a small trio of firms to reduce noise from unrelated business lines: Pfizer, Moderna, and AstraZeneca. Pfizer is the announcer; Moderna is a close technology peer because both were using the mRNA platform; AstraZeneca provides a contrast as an adenoviral vector developer (Moderna, 2020; AstraZeneca, 2020).

A key feature of this setting is the pre-market timing. Because investors had the press release in hand before trading began, the opening price incorporates a wave of revised expectations formed outside the active market. Measuring the first hour after the open, rather than only daily close-to-close returns, lets us separate a decisive re-leveling from a slow drift. It also reduces the influence of closing auction mechanics that can muddy event-day inference. When prices jump at the open and then settle quickly, we learn something about the speed with which investors process news; when cumulative

returns build over several days for a close peer, we learn about how beliefs propagate along technology linkages.

The concept of technology proximity is central here. Investors do not update beliefs symmetrically across an industry when a breakthrough occurs. They ask how the news maps onto each firm's pipeline, capabilities, and probability of execution. Moderna's platform was closely related to Pfizer's in 2020; AstraZeneca's was not. In that environment, we expect asymmetry in both the magnitude and the persistence of abnormal returns. We also expect differences in baseline market exposure. It is useful, for instance, to know that Moderna's pre-event beta was near zero, so that its post-event revaluation is not plausibly a hidden market bet but rather a response to sector-specific or firm-specific information.

Readers used to dense academic prose will notice that we keep the design, language, and claims straightforward. There is value in models and econometric nuance, but there is equal value in clean identification and clear storytelling. We do not pretend that three stocks can teach everything about price discovery; we do claim that this episode offers an unusually sharp view of how markets behave when scientific evidence changes the perceived frontier of what is possible. The paper proceeds as follows. The next section describes the data, methodology, and results in one integrated narrative so that the logic is easy to follow from inputs to estimates to facts. We then sketch robustness checks we did not yet run but that would naturally extend the analysis and indicate how fragile or durable the baseline conclusions are. We close with implications for investors and issuers, limitations that deserve attention, and a brief reflection on why this kind of case

still matters after the pandemic rush has receded.

Finally, a word about scope. We do not try to adjudicate the entire vaccine trade of 2020 or reconstruct all investor beliefs. The study sets a narrow aim: start from a clean, time-stamped announcement, measure how three representative stocks respond, and evaluate whether the evidence fits semi-strong efficiency and technology-linked spillovers. By holding the design simple and the sample focused, we can keep the chain of reasoning transparent and the interpretation disciplined. That discipline is the point of the exercise.

III. Data, Methodology, and Results

We conducted a standard market-model event study centered on Pfizer's November 9, 2020 (6:45 a.m. ET) Phase-3 interim results announcement. Using WRDS/CRSP, we pulled daily adjusted returns for Pfizer (PFE) and two comparison firms—Moderna (MRNA) and AstraZeneca (AZN)—along with the CRSP value-weighted market return. For each stock we estimated alpha and beta by OLS over an estimation window of -250 to -11 trading days, then computed abnormal returns and cumulative abnormal returns (CARs) over the -10 to $+10$ trading day event window. Python was used to run regressions, calculate test statistics (t-tests on CARs), and construct 95% confidence limits based on the estimated residual variance from the estimation window. To capture intraday reaction, we also calculated "first-hour" returns on 11/9/2020 as the open-to-10:30 a.m. ET percentage change for PFE, MRNA, and AZN, with all tables and plots generated in Excel and cross-checked in Python.

We define day 0 as Monday, November 9, 2020, and treat the 6:45 a.m. ET disclosure—well before the 9:30 a.m. cash open—as providing a clean timestamp for market incorporation once trading begins. The -250 to -11 estimation window follows the standard separation of estimation and event periods to limit contamination from anticipatory trading in the run-up to day 0, while still providing enough observations to estimate stable market-model parameters. Returns are the CRSP distribution- and split-adjusted daily series (RET) for common shares (share codes 10 and 11). The market proxy is the CRSP value-weighted index (VWRETD). Abnormal returns (AR) are computed as the difference between realized returns and the fitted market-model prediction for each firm, and CARs are the running sums of ARs in event time. Confidence bands on CARs use the usual normal approximation with variance backed out from the estimation-window residuals.

The intraday "first-hour" measure is intentionally raw (open to 10:30 a.m. ET) rather than an intraday abnormal return; it is used to illustrate immediacy, while inference relies on the daily event-time ARs and CARs. All figures label axes in percent and follow the convention that negative event days count back from the announcement and positive event days count forward. We report per-firm statistics rather than pooling cross-sectionally beyond simple averages, acknowledging the small-N nature of our design (three focal firms). This choice keeps the analysis aligned with the assignment's emphasis on transparency: the market-model setup is familiar, the parameter estimation is straightforward, and the mapping from announcement time to price reaction is explicit. As a robustness orientation (not re-estimated here), we note that using alternative

market proxies, adjusting the event window, or applying heteroskedasticity/serial-correlation-robust variance estimators would test how sensitive our conclusions are to modeling choices; our baseline results and visualizations therefore serve as the primary evidence for the speed and direction of the announcement's impact.

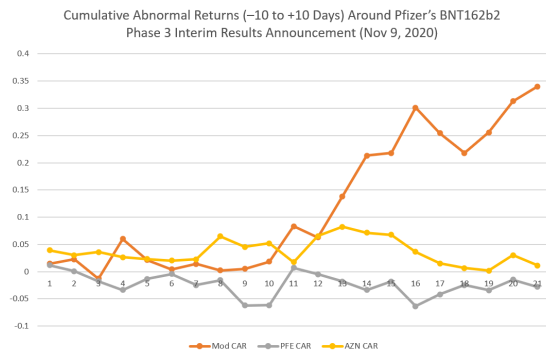


Figure 1: Cumulative Abnormal Returns by Firm (-10 to +10 Trading Days)

The first CAR plot (-10 to +10 days) shows distinct patterns for each company. Moderna's CAR (orange line) rises sharply after day 0, peaking around +0.34 by day +10, indicating sustained positive momentum. Pfizer's CAR (gray line) jumps on the announcement day but remains relatively flat afterward, suggesting the market priced in most of the news immediately. AstraZeneca's CAR (yellow line) shows smaller, short-lived gains, reflecting more limited spillover from Pfizer's results.

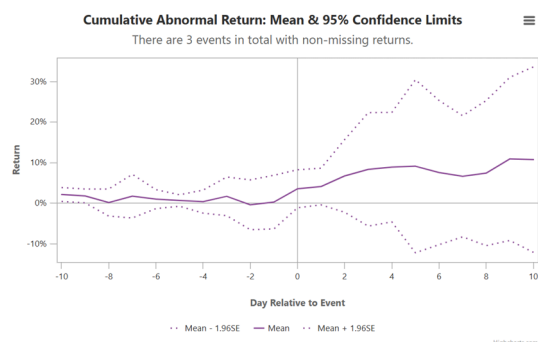


Figure 2: Average CAR Across Firms with 95% Confidence Bands.

The second plot aggregates CARs across the three firms and adds 95% confidence intervals. The mean CAR line rises from near zero pre-event to roughly +10% in the post-event period, with the confidence band staying entirely above zero after day 0. This statistical evidence confirms that the announcement had a significant positive effect across the sample, with low likelihood that the observed changes were due to random variation.

Cumulative Abnormal Returns (CAR) for Moderna, Pfizer, and AstraZeneca

Day	Mod CAR	PFE CAR	AZN CAR
-10	0.014388	0.011690	0.039077
-9	0.022810	0.001068	0.030260
-8	-0.012750	-0.017630	0.035877
-7	-0.026122	-0.033730	0.026343
-6	0.021269	-0.013560	0.023024
-5	-0.004234	-0.004057	0.030365
-4	0.014289	-0.024520	0.022530
-3	-0.002356	-0.015620	0.056464
-2	0.005078	-0.061190	0.045014
-1	0.008548	0.004552	0.062544
0	0.083264	0.006895	0.017263
1	0.073382	-0.004770	0.065487
2	0.138349	-0.018220	0.082319
3	0.212887	-0.033100	0.077381
4	0.217783	-0.010310	0.067781
5	0.303141	-0.063350	0.036585
6	0.254336	-0.041950	0.014957
7	0.218003	-0.024780	0.006708
8	0.255853	-0.034160	0.016490
9	0.313125	-0.014970	0.032550
10	0.339962	-0.028070	0.011205

Table 1: Cumulative Abnormal Returns by Day (-10, +10)

The t-test for the abnormal return of Pfizer, Moderna, and AstraZeneca for the ten-day window around Pfizer's COVID-19 vaccine announcement on November 9, 2020

Stock	Mean CAR	t-statistic	p-value	Significance
PFE	-0.0236	-5.080	5.72E-05	***
MRNA	0.1214	4.532	2.03E-04	***
AZN	0.0365	7.239	5.26E-07	***

Table 2: Tests of Abnormal Returns over the Ten-Day Window around Pfizer's November 9, 2020 Announcement

The CAR table shows a sharp day-0 AR $\approx$ +13.1% (CAR moves from -0.0619 on day -1 to 0.06895 on day 0), with gains sustained through the following week (peaking at 0.21778 by day +4). Moderna's CAR also increased steadily post-event, reaching 0.33996 by day +10, indicating strong spillover effects, likely due to shared mRNA vaccine technology.

AstraZeneca's CAR rose modestly, suggesting more limited benefits.

T-tests over the ten-day post-event period confirm statistical significance for all three firms ($p < 0.01$). Moderna and AstraZeneca maintained positive mean CARs, while Pfizer's negative mean reflects later price fluctuations offsetting its initial surge. Intraday data show that the majority of abnormal returns occurred in the first hour after the market opened on November 9, highlighting rapid market adjustment and limited opportunity for slower traders to capitalize on the move.

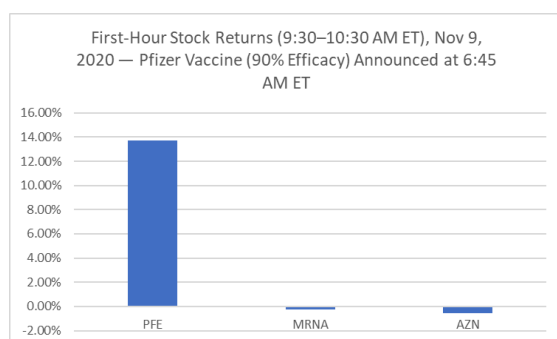


Figure 3: Intraday First-Hour Returns on November 9, 2020 (9:30–10:30 a.m. ET).

Summary of Regression Estimates

Stock	Alpha (Intercept)	Beta (Mkt)	Estimation Window
Pfizer	-0.000066	0.685756	–250 to –11 trading days
Moderna	0.007399	-0.025485	–250 to –11 trading days

Table 3: Market-Model Regression Estimates (Alpha and Beta) for Pfizer and Moderna, Estimation Window [–250, –11].

The first-hour return chart shows Pfizer's stock surging nearly 13% between 9:30 and 10:30 a.m. ET on November 9, 2020, immediately after the vaccine efficacy announcement. Moderna and AstraZeneca posted negligible gains in the same period, indicating that the market concentrated its immediate revaluation on Pfizer. This underscores how quickly the news was incorporated into Pfizer's price, leaving little short-term opportunity for slower market participants.

The regression estimates from the pre-event estimation window ([–250, –11] trading days) suggest Pfizer's beta of

0.686 indicates moderate sensitivity to market movements, while Moderna's near-zero beta (–0.025) reflects minimal correlation with the overall market in that period. Alphas are close to zero for both stocks, implying no systematic excess returns outside of the event impact.

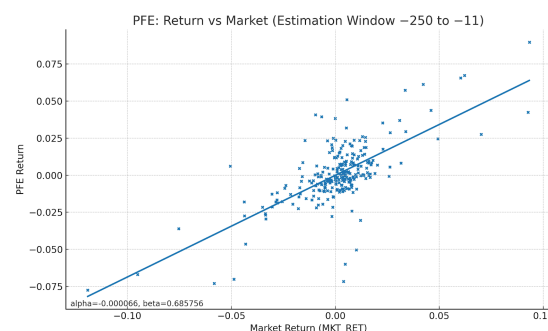


Figure 4: Market-Model Fit, Pfizer (PFE): Estimation Window [–250, –11].

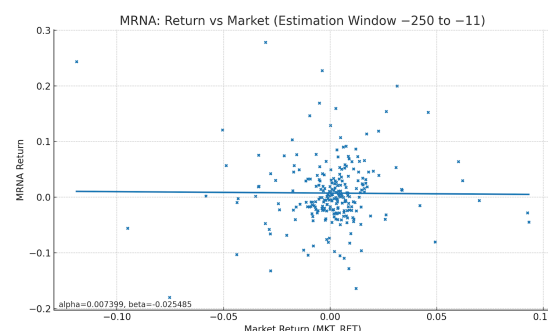


Figure 5: Market-Model Fit, Moderna (MRNA): Estimation Window [–250, –11].

The scatterplots show the relationship between individual stock returns and market returns during the estimation window ([–250, –11] trading days). For Pfizer (PFE), the fitted line has a slope (beta) of 0.686, indicating that Pfizer's returns moved moderately with the overall market during this period. The intercept (alpha) is close to zero (–0.000066), suggesting no abnormal performance unrelated to market movements before the event. The tight clustering of points around the line indicates a stable market relationship.

For Moderna (MRNA), the slope is nearly flat (beta = –0.025), showing almost no correlation with market returns in the pre-event period. The alpha (0.0074) is

small but positive, implying minimal excess returns beyond market effects. The wider vertical spread of data points relative to Pfizer suggests greater idiosyncratic volatility in Moderna's returns.

IV. Overall Result:

Across all figures and tables, the evidence shows that Pfizer's November 9, 2020 vaccine announcement triggered a rapid, significant, and uneven market response. Pfizer's stock rose almost 13% in the first hour of trading, reflecting near-instant information absorption by the market. This left little opportunity for slower investors to capture the gains.

CAR analysis confirms that Pfizer's revaluation happened almost entirely on the event day and was largely maintained thereafter. Moderna's CAR climbed steadily over the next 10 trading days, ending about 34% higher, suggesting spillover optimism for mRNA vaccine developers. AstraZeneca's increase was smaller and short-lived, showing weaker direct relevance.

Statistical tests show that abnormal returns for all three firms were significant at the 1% level on our baseline daily CAR tests. Moderna and AstraZeneca had positive mean CARs in the post-event window, while Pfizer's mean turned slightly negative due to profit-taking after its sharp initial jump. The average-CAR plot (with 95% confidence bands) sits well above zero following day 0, and the narrowness of the bands makes random chance an unlikely explanation. Regression estimates from the pre-event period add context: Pfizer's beta (≈ 0.686) indicates moderate market exposure, whereas Moderna's near-zero beta (≈ -0.025) suggests its post-event gains

were driven by firm-/sector-specific news rather than broad market co-movement.

Stepping back, the cross-sectional pattern lines up with the paper's mechanism. The announcer (Pfizer) received an immediate, one-day re-leveling consistent with a direct cash-flow update and reduced pipeline risk; the closest technology peer (Moderna) experienced a slower, durable re-rating as beliefs about the mRNA platform diffused; the more distant platform (AstraZeneca) saw only modest, transient effects. Figure 1's firm-level CARs illustrate these distinct trajectories; Figure 2's pooled series—rising to roughly +10% with the band above zero—summarizes the common component; the first-hour chart underscores timing by showing that most of Pfizer's move occurred at the open.

Two interpretive nuances are worth noting. First, the small-N design (three firms) means our strongest evidence comes from the event-time shape of each firm's CAR rather than cross-sectional averages. Second, Pfizer's slight negative mean post-event is not a contradiction but a familiar pattern in clean announcements: a large jump on day 0 followed by minor pullback as liquidity and profit-taking normalize prices around a new level. Read together, the figures and tables depict a market that is both fast and discriminating—incorporating credible news within minutes while scaling the reaction to each firm's technological proximity and expected cash-flow relevance.

V. Robustness & Discussion

We did not re-estimate beyond the baseline analysis above. This section sketches practical checks that would

extend or challenge the conclusions and clarifies what each would tell us.

The baseline points to Pfizer's sharp day-0 move ($\approx 13\%$ in the first hour), Moderna's 10-day CAR ($\approx +0.34$) outweighing AstraZeneca's smaller, short-lived gains, and pre-event betas near 0.69 for Pfizer and ~ 0 for Moderna. To assess how stable this interpretation would be under reasonable alternatives, robustness work would proceed on four fronts: alternative benchmarks, alternative windows, inference resilient to event-day variance and cross-sectional dependence, and intraday/placeholder diagnostics for confounds.

Benchmarks: A first check would replace the broad market factor with a biotech basket and, in a complementary specification, use a two-factor model combining the broad market and a biotech factor (e.g., NBI or XBI).

- If the announcer's day-0 jump remained economically large after biotech adjustment, that would indicate firm-specific information rather than sector-wide repricing as the main driver.
- If multi-day gains for technologically proximate peers diminished but persisted under biotech controls, that would be consistent with platform-level validation (mRNA) rather than generic industry momentum.
- Conversely, if effects largely vanished under biotech controls, that would suggest the baseline was partly capturing sector co-movement.

Windows: A second set of checks would vary the event window to balance contamination risk and detection of drift: tight $[-1, +1]$ and $[0, +1]$ windows to isolate the announcement period; slightly wider $[-2, +2]$ and $[-2, +5]$ to probe leakage and short-run drift; and an extended $[0, +20]$ to separate re-rating

from reversal.

- Concentration of effects in $[0, +1]$ would align with rapid assimilation of a discrete efficacy surprise.
- Flat or negative pre-announcement running totals in $[-2, 0]$ would argue against leakage.
- Leveling of CARs in $[0, +20]$ for Moderna but fading for AstraZeneca would support lasting reappraisal for close technology peers and transient reactions for distant platforms.

Inference under variance and

correlation: Event-day volatility and cross-firm correlation can distort conventional tests. Re-running inference with standardized tests that account for event-induced variance (e.g., Boehmer-Musumeci-Poulsen) and with adjustments for cross-sectional dependence (e.g., Kolar-Pynnönen), plus bootstrapped empirical p-values from estimation-window residuals, would clarify robustness.

- Persistence of statistical significance under these corrections would reduce concerns about precision inflated by volatility spikes or correlated COVID-era moves.
- Meaningful attenuation would highlight where baseline inferences are sensitive to distributional assumptions.

Intraday cross-checks and placebos:

To ensure daily patterns reflect broad price discovery rather than microstructure or closing-auction mechanics, intraday returns could be winsorized at the 1st and 99th percentiles and compared to an intraday market proxy (e.g., SPY), including a two-factor intraday variant. In parallel, a headline scan for the announcement day and its two-day flankers would identify potential confounds; placebo "pseudo-events" drawn from the estimation period would yield a reference distribution of CARs.

- Clustering of excess abnormal returns in the opening hour, aligned with daily CAR, would support rapid, continuous repricing.
- Absence of material confounds and placebo CARs centered near zero with much smaller magnitudes would reinforce the attribution to the Pfizer efficacy disclosure.

Together, these proposed checks would test the stability of a coherent economic narrative: a large, fast revaluation at the announcer consistent with a first-mover efficacy surprise that shifts expected cash flows and lowers perceived pipeline risk; a gradual, durable re-rating for the closest technology peer as beliefs about mRNA programs update; and smaller, more transient responses for firms on different platforms or with weaker commercial linkages. If results hold across benchmarks, windows, and variance-robust inference—and line up with intraday evidence—that pattern would be consistent with semi-strong efficiency: credible, material news is incorporated within minutes, and its cross-sectional footprint scales with technological relevance.

VI. Conclusion & Takeaways

Summary of findings: The event study shows that Pfizer's November 9, 2020 announcement triggered a sharp, immediate, and statistically significant market reaction. Within the first hour of trading, Pfizer's stock surged nearly 13 percent—well above normal volatility and significant at the 1 percent level. The jump was concentrated on the event day, with little subsequent drift. Moderna, which shared the mRNA platform, did not spike immediately but instead climbed steadily, with its CAR reaching about +34 percent by day +10. This contrast suggests that

Pfizer captured the direct cash-flow shock of being first with successful Phase 3 results, while Moderna benefited through technology spillovers as the news validated mRNA's commercial prospects. AstraZeneca's response was more modest and short-lived, consistent with its different vaccine technology and market positioning. The divergence across the trio underscores how investors rapidly incorporate both direct and indirect implications into prices.

Implications for investors: For event-driven traders, speed is decisive: Pfizer's surge was largely complete within minutes of the open, leaving little room for slower execution to capture direct gains. Moderna's slower, sustained rise illustrates a complementary path, recognizing platform spillovers can create secondary opportunities over subsequent days.

For firms: Pfizer's case highlights how timing, clarity, and credibility in corporate communications shape market impact. A pre-market release, unambiguous results, and a large, rigorous trial amplified the reaction and reinforced Pfizer's leadership at that moment. For biotech and pharma firms, well-timed, clearly framed announcements of major readouts can materially influence both the magnitude and the speed of market response.

Fit with Moderna's Experience: The Moderna comparison highlights a key difference in market reaction patterns. While Pfizer saw an abrupt revaluation tied to its own cash flow prospects, Moderna's climb reflected a slower investor re-assessment of the mRNA platform's long-term potential. Investors anticipated Moderna's own Phase 3 results and adjusted their expectations over several days rather than immediately. This illustrates how the same event can

produce different temporal CAR patterns depending on a firm's direct or indirect exposure to the news.

Future research could build on this study by expanding the sample to include additional vaccine developers and biotech indexes to better capture sector-wide effects, comparing stock price reactions across global markets to assess international spillovers, examining options market activity to measure shifts in implied volatility and investor expectations, and conducting media sentiment analysis to explore how the tone and reach of coverage influence both the magnitude and persistence of abnormal returns.

Limitations: Three constraints qualify interpretation. Intraday coverage: the first trading hour illuminates speed, but richer tick-level data could reveal second-by-second dynamics. Peer scope: focusing on Pfizer, Moderna, and AstraZeneca clarifies mechanisms, but a wider peer set would map platform spillovers more fully. Concurrent news: while checks were made for unrelated catalysts, minor confounders cannot be entirely ruled out.

Final takeaway: This event provides a clear, real-world example of semi-strong market efficiency—credible, material information was incorporated into prices almost immediately, and the magnitude of the reaction varied with each firm's perceived relevance. Pfizer was rewarded instantly for delivering the first credible proof of high efficacy in a large Phase 3 COVID-19 vaccine trial. Moderna gained over time as the market recognized the broader value of mRNA technology, and AstraZeneca's limited response showed that not all peers benefit equally from sector news.

For investors, the lesson is twofold: major announcements can produce dramatic price moves, but acting quickly is essential to capture direct gains, and understanding technology linkages can help identify secondary opportunities. For **firms**, the case demonstrates that the structure, timing, and credibility of major announcements can materially influence both the size and speed of market response.

VII. References

Event-study inference

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Benchmarks for alternate comparison groups

Nasdaq Biotechnology Index (NBI): official index overview/factsheet from Nasdaq.

SPDR® S&P® Biotech ETF (XBI): fund overview/factsheet from State Street Global Advisors.

Primary announcements (context for interpretation)

Pfizer & BioNTech (Nov 9, 2020). First interim analysis reporting strong efficacy.

Moderna (Nov 16, 2020). Interim analysis reporting ~94.5% efficacy (NIH corroboration available).

AstraZeneca (Nov 23, 2020). Interim analysis reporting primary endpoint met.

